Week 13 Notes

STAT 251 SECTION 01

Comparing two means

Last week we learned how to compare two populations across a qualitative response variable by comparing their relative proportions. Now we will learn how we can compare two groups on a quantitative response variable by comparing their means. There are many cases where researchers might be interested in comparing groups across a quantitative variable. For example, a teacher may be interested in comparing the average test scores of students in two different classes (Class A and Class B). A social scientist may be interested in whether there is a difference in salaries between men and women within a company. In any case, we are again dealing with two samples of sizes n_1 and n_2 which come from populations *population 1* and *population 2* and with means as μ_1 and μ_2.

Population	Population Mean	Population Stdev	Sample Size	Sample Mean	Sample Stdev
1	μ_1	σ_1	n_1	$\bar{x}_1$	s_1
2	μ_2	σ_2	n_2	$\bar{x}_2$	s_2

Let $\mu_d = \mu_1 - \mu_2$ be the mean difference between the two populations. We can construct confidence intervals to estimate the average difference as well as conduct hypothesis tests to determine if two populations are significantly different.

A confidence interval for the mean difference μ_d

We will begin by assuming that we are dealing with two independent samples from two distinct populations. The natural estimator of μ_d is the difference between the sample means

$$\hat{\mu}_d = \bar{x}_1 - \bar{x}_2$$

To base our inference on the above estimator, we must know its sampling distribution. If we assume that both samples come from populations that are normally distributed, then the distribution of the difference will also be normally distributed and the variances will add:

$$Var[\mu_1 - \mu_2] = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

In practice, σ_1^2 and σ_2^2 are unknown population parameters but can be estimated by substituting in the sample variances s_1^2 and s_2^2. This gives the following standard deviation for the estimator $\hat{\mu}_d$

$$SE(\hat{\mu}_d) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Just as it was for a single mean, estimating the population variances with the sample variances adds additional uncertainty. Thus, the sampling distribution for the difference in two means is t-distributed, although the degrees of freedom are calculated differently. The actual formula for the degrees of freedom is quite messy and difficult to compute:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1-1}\left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2-1}\left(\frac{s_2^2}{n_2}\right)^2}$$

We will instead compute the degrees of freedom by taking the smaller of $n_1 - 1$ and $n_2 - 1$ which is a lower bound on the formula above

$$\frac{\hat{\mu}_d - \mu_d}{SE(\hat{\mu}_d)} \sim t(\min(n_1 - 1, n_2 - 1))$$

The margin of error for estimating the mean difference at the $100(1-\alpha)\%$ level is given by

$$m = t_{1-\alpha/2} \times SE(\hat{\mu}_d)$$

which gives the following confidence interval for the mean difference

$$(\bar{x}_1 - \bar{x}_2) \pm t_{1-\alpha/2} \times SE(\hat{\mu}_d)$$

Example: Comparing Customer Satisfaction Levels - Suppose we want to compare the customer satisfaction levels of two competing cable television companies, Company 1 and Company 2. We randomly select customers from each company and ask them to rate their cable companies on a five-point scale (with 1 being least satisfied and 5 most satisfied). The data are summarized in the table below:

Group	Number of customers	Sample Mean Rating	Sample Stdev
Company 1	174	4.2	0.8
Company 2	355	3.8	1.1

Estimate the difference in mean satisfaction levels between the two companies at the 95% confidence level.

Let μ_1 represent the mean satisfaction score for company 1 and μ_2 represent the mean satisfaction score for company 2. The estimated mean difference in customer satisfaction is

$$\hat{\mu}_d = 4.2 - 3.8 = 0.4$$

Since the confidence level is 95% the standard score will be approximately the $t_{1-0.05/2} = t_{0.975}$ or the $97.5th$ percentile of a t-distribution with 173 degrees of freedom. Since the t and z distributions are approximately equal for sample sizes ≥ 30 we will use the standard score 1.96. The standard error of the difference is given by

$$SE(\hat{\mu}_d) = \sqrt{\frac{0.8^2}{174} + \frac{1.1^2}{355}} = 0.0842$$

Which yields the follow margin of error

$$m = 1.96(0.0842) = 0.165$$

The 95% confidence interval is $[0.24, 0.56]$. Thus, at the 95% confidence level we estimate the mean difference in customer satisfaction score to be no less than 0.24 and no more than 0.56. Notice that the above confidence interval does not span zero and thus likely constitutes a real, but small, difference between the companies

A significance test for comparing two means: independent samples

We can also test for a difference between two groups on a quantitative response variable by comparing their means. The set up for a significance test for a difference in two means is more or less the same as we saw previously for two proportions, however, the calculations now involve $\bar{x}_1$, $\bar{x}_2$, s_1 and s_2. Recall that, under the assumption of normality for both populations and when using the sample standard deviations to estimate σ_1 and σ_2 the difference in means is t-distributed. We will see that test statistic t_{obs} is the standardized difference and has approximately the same distribution. Here's how we set up a two-sample hypothesis test for means:

1. Assumptions

- Data constitute a simple random sample
- The two groups are independent
- Both populations are normally distributed

2. Null and Alternative Hypotheses

- The **null** hypothesis is generally that the two population means are equal (i.e no difference)

$$H_0 : \mu_1 = \mu_2 \quad \text{which is the same as} \quad H_0 : \mu_1 - \mu_2 = 0$$

- The alternative hypothesis can be either lower, upper, or two tailed just as it was with our univariate tests. The table below gives the possible alternative hypotheses and their critical values:

Alternative Hypothesis	Critical Value	Rejection Region
$H_A : \mu_1 - \mu_2 \neq 0$	$t_{1-\alpha/2}$	$\lvert t \rvert \geq t_{1-\alpha/2}$
$H_A : \mu_1 - \mu_2 < 0$	t_{α}	$t \leq t_{\alpha}$
$H_A : \mu_1 - \mu_2 > 0$	$t_{1-\alpha}$	$t \geq t_{1-\alpha}$

3. Test statistic

- Under the null hypotheses the test statistic is the standardized difference

$$t_{obs} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

The above test statistic is only approximately t-distributed. However, the lower bound for the degrees of freedom given by $\min(n_1 - 1, n_2 - 1)$ generally leads to conservative estimates and tests and is thus favorable.

4. p-value

- The p-values corresponding to each alternative hypothesis are listed in the table below

Alternative Hypothesis	p-value
$H_A : \mu_1 - \mu_2 \neq 0$	$P(\lvert t \rvert \geq \lvert t_{obs} \rvert \lvert H_0)$
$H_A : \mu_1 - \mu_2 < 0$	$P(t \leq t_{obs} \lvert H_0)$
$H_A : \mu_1 - \mu_2 > 0$	$P(t \geq t_{obs} \lvert H_0)$

5. Decision rule

$$\text{Reject } H_0 \text{ if } p\text{-value} < \alpha$$

The pooled t test

There is one situation where the above test statistic has exactly a t-distribution and that occurs when we can assume that both populations have the same standard deviation. In this case, we need only substitute a single sample standard deviation for σ_1 and σ_2. We call this the pooled estimate of the standard deviation:

$$s_{pooled} = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$$

s_{pooled} is called the **pooled estimator** of the common standard deviation σ because it combines the information from both samples.

The test statistic under the assumption of equal variances becomes

$$t_{obs} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_{pooled}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

which is exactly distributed as a t distribution with $n_1 + n_2 - 2$ degrees of freedom. The test based on the pooled estimate of the standard deviation is more powerful because the sampling distribution is exact and not an approximation as it is in the unpooled case.

Which test should you use?

- In practice, a significance test for determining equality of variances can be run prior to the t-test to determine if the pooled or unpooled test statistic should be used. More information about this test can be found in the NIST/SEMATECH e-Handbook of Statistical Methods: F-Test for Equality of Two Variances (https://www.itl.nist.gov/div898/handbook/eda/section3/eda359.htm#:~:text=An%20F%2Dtest%20(Snedecor%20and,the%20variances%20are%20not%20equal.)

Example: NCAA Women's Basketball - Sunday April 7th South Carolina and Iowa contended for the NCAA women's basketball championship. Consider the following table which gives the opponent and number of points scored for both teams in their last 15 games leading up to the championship.

South Carolina	Matchup	Iowa	Matchup
78	NC State	71	UConn
70	Oregon State	94	LSU
79	Indiana	89	Colorado
88	UNC	64	West Virginia
91	Presbyterian	91	Holy Cross
79	LSU	94	Nebraska
74	Tennessee	95	Michigan
79	Texas A&M	95	Penn State
76	Tennessee	93	Ohio State

South Carolina	Matchup	Iowa	Matchup
98	Arkansas	108	Minnesota
103	Kentucky	101	Illinois
72	Alabama	69	Indiana
70	Georgia	106	Michigan
66	Tennessee	79	Nebraska
83	UConn	111	Penn State

The summary of the last 15 games is given below:

Population	Population Mean	Sample Size	Sample Mean	Sample Stdev
Iowa	μ_1	$n_1 = 15$	90.67	14.21
South Carolina	μ_2	$n_2 = 15$	80.40	10.57

Conduct an unpooled two sample t-test at the $\alpha = 0.05$ significance level to determine if there is a significant difference between the championship teams in the mean number of points scored per game.

1. Let μ_1 be the mean number of points scored per game for Iowa and μ_2 be the mean number of points scored per game for South Carolina. The null hypothesis is that there is no difference in the mean number of points between the two teams

$$H_0 : \mu_1 = \mu_2$$

The alternative hypothesis is that the two teams are different

$$H_A : \mu_1 \neq \mu_2$$

2. The estimated difference is given by $\hat{\mu}_d = \bar{x}_1 - \bar{x}_2 = 10.27$. The test statistic for the unpooled test is

$$t_{obs} = \frac{90.67 - 80.40}{\sqrt{\frac{14.21^2}{15} + \frac{10.57^2}{15}}} \approx 2.246$$

which is approximately distributed as

$$t_{obs} \sim t(df = 14)$$

3. The p-value is

$$p\text{-value} = 2\left[P(|t| \geq |t_{obs}||H_0)\right] = 2(0.0207) = 0.0414$$

4. Based on the p-value above we would narrowly reject the null hypothesis and conclude that there is a significant difference in the mean number of points scored per game between Iowa and South Carolina. The outcome of the championship was an 87 - 75 victory for South Carolina. Given the conclusion of our test, this is a testament to South Carolina's defensive prowess in being able to shut down a high scoring team like Iowa.

Paired t test

Comparative methods (i.e methods that compare two groups/samples across a quantitative variable) are more common and often preferred over one sample procedures. One common comparative design that makes use of single-sample procedures for a population mean is called the paired t-test. This test is used to draw inferences about a population mean in matched-pairs designs and can be thought of as the parametric analogue of the *sign test* we learned previously. Recall that in matched-pairs studies subjects are matched based on common characteristics and the quantitative outcome is compared within each pair.

- matched-pairs designs are most common when randomization is not possible
- often used when observations are taken on the same subject (before and after treatment)

In a paired t-test we convert a comparative study into a one-sample procedure by making the response variable the difference between the two observations in each pair $d_i = x_i - y_i$. The parameter under study is the population mean difference μ_d which can be estimated by the sample mean difference $\bar{x}_d = \bar{x} - \bar{y}$ computed over all pairwise differences. The standard deviation of the difference σ_d is then estimated with the sample standard deviation of the difference. Inference is then treated in the same way as one-sample procedures for a population mean

- the $100(1-\alpha)\%$ confidence interval for the mean pairwise difference is

$$\bar{x}_d \pm t_{1-\alpha/2}\frac{s_d}{\sqrt{n}}$$

- the test statistic for a significance test regarding the pairwise difference is given by

$$t_{obs} = \frac{\bar{x}_d - \mu_d}{s_d/\sqrt{n}} \sim t(n-1)$$

- by treating the above procedure as a one-sample t test we make the assumption that the population distribution of differences is approximately normal.

Example: Revisiting the Spousal Dominance Problem - Recall the study comparing the ratings of husbands and wives on the perceived relative influence of each member of the couple on a major financial decision.

Couple	Husband	Wife	Difference
1	5	3	2
2	4	3	1
3	6	4	2
4	6	5	1
5	3	3	0
6	2	3	-1
7	5	2	3
8	3	3	0
9	1	2	-1
10	4	3	1
11	5	2	3

Couple	Husband	Wife	Difference
12	4	2	2
13	4	5	-1
14	7	2	5
15	5	5	0
16	5	3	2
17	5	1	4

A summary of the difference in rating between husbands and wives is given below:

Parameter	Estimate	Stdev
μ_d	1.35	1.77

Conduct a paired t-test at the $\alpha = 0.05$ significance level to determine if husbands and wives differ in their perceived influence on financial decision making

Summary of inference procedures

The table below summarizes the procedures we have learned so far in the course

Parameter	Type	Response Variable	$100(1-\alpha)\%$ CI	Test statistic	Distribution
p	one-sample	Categorical	$\hat{p} \pm Z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$\frac{\hat{p}-p_0}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$	z
$p_1 - p_2$	two-sample	Categorical	$(\hat{p}_1 - \hat{p}_2) \pm Z_{1-\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	$\frac{\hat{p}_1-\hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$	z
μ	one-sample	Quantitative	$\bar{x} \pm t_{1-\alpha/2,n-1}\frac{s}{\sqrt{n}}$	$\frac{\bar{x}-\mu_0}{s/\sqrt{n}}$	$t(n-1)$
$\mu_1 - \mu_2$	two-sample	Quantitative	$(\bar{x}_1 - \bar{x}_2) \pm t_{1-\alpha/2,\min(n_1-1,n_2-1)}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	$\frac{\bar{x}_1-\bar{x}_2}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}}$	$t(\min(n_1-1, n_2-1))$
$\mu_1 - \mu_2$	pooled two-sample	Quantitative	$(\bar{x}_1 - \bar{x}_2) \pm t_{1-\alpha/2,n_1+n_2-2}s_{pooled}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$	$\frac{\bar{x}_1-\bar{x}_2}{s_{pooled}\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}$	$t(n_1+n_2-2)$
μ_d	matched pairs	Quantitative	$\bar{x}_d \pm t_{1-\alpha/2,n-1}\frac{s_d}{\sqrt{n}}$	$\frac{\bar{x}_d-\mu_d}{s_d/\sqrt{n}}$	$t(n-1)$

Statistical Inference for qualitative data

Despite significantly expanding our statistical tools for inference, we are still quite limited in our ability to analyze qualitative responses. Our tests for a proportion are useful for comparing a single category of a qualitative variable across one or two samples, but what if we are interested in comparing multiple categories across more than one population? We will now explore some of the common tools for analyzing qualitative data. The inference techniques for analyzing qualitative variables are generally concerned with comparing a response variable with two or more categories

across an explanatory variable with two or more groupings by exploring how the conditional distribution of the response varies across the groups of the explanatory variable. In other words, these inference procedures are designed to analyze the statistical association between two multi-category qualitative variables.

Such problems are typically represented as a table where the rows represent the groupings of the explanatory variable and columns represent the categories of the response variable - often referred to as “two-way tables”.

The Chi-square goodness of fit test

Like we have done with means and proportions, we will begin with a one-sample procedure for comparing the distribution of a categorical variable to some hypothesized model. This procedure is called the χ^2 (“chi-squared”) goodness of fit test because it used to determine how well the observed distribution fits some proposed model of the data. We will illustrate this idea by starting with an example:

Example: Gregor Mendel’s Peas - Gregor Mendel conducted a study in which he crossed pea plants. The plants he crossed had the alleles (i.e., a form or version of a gene) for both yellow Y and green peas y — a trait controlled by a single locus. He believed two of his laws applied here: the Law of Segregation and the Law of Dominance. These laws would determine the probabilities of certain genotypes. This can be summarized by a Punnett square

Yellow	Green
$YY = 0.25$	$Yy = 0.25$
$Yy = 0.25$	$yy = 0.25$

- If Mendel’s Laws are true, what do they imply about the probability distribution for these four outcomes (i.e., YY, Yy, yY, or yy)? And what does this imply about the probability distribution of color (i.e., yellow or green) ?

Mendel bred 8023 offspring. Of these, 6022 were yellow, and 2001 were green. Do his laws “fit” the data? That is, does the trait of color in pea plants exhibit what we would call Mendelian inheritance?

Color	Expected Frequency	Observed Count	Observed Frequency
Yellow	0.75	6022	0.7506
Green	0.25	2001	0.2494

The above problem amounts to comparing the observed frequencies of each color to the frequencies that we would expect under the Mendelian model of inheritance. We can test whether the data fit Mendel’s model by comparing the observed frequencies of pea plant colors to the frequencies we would expect assuming Mendelian Inheritance and Dominance.

In this case the null hypothesis is that the offspring of the pea plants follows Mendelian inheritance or

$$H_0 : p_{yellow} = 0.75; p_{green} = 0.25$$

The alternative is that one or more of the frequencies for each color do not follow the frequencies defined in the null hypothesis. To generalize for a categorical variable with k possible outcomes the null becomes $H_0 : p_i = c_i; \forall i \in [1 : k]$ and H_A : at least one $p_i \neq c_i$.

The above hypothesis test uses a χ^2 test statistic defined as

$$X^2 = \sum_{i=1}^{k} \frac{(\text{observed count}_i - \text{expected count}_i)^2}{\text{expected count}_i}$$

- k is the number of possible outcomes
- the observed count is the number of observations we observed with outcome i
- the expected count is the number of observations we expect to have outcome i which is calculated under the null hypothesis as np_i
- We can think of this calculation as measuring the total (relative) squared distance from the observed model to the model we defined under the null hypothesis.

X^2 can take on values between 0 and positive ∞. A value of zero means the observed model exactly follows the expected model. Values greater than zero indicate a deviation from the expected model. Large values of X^2 indicate evidence against the null hypothesis.

Referring back to Mendel's pea plants we compute the expected number of yellow pea plants as $8023 \times 0.75 = 6017.25$ and the expected number of green pea plants to be $8023 \times 0.25 = 2005.75$. The test statistic is

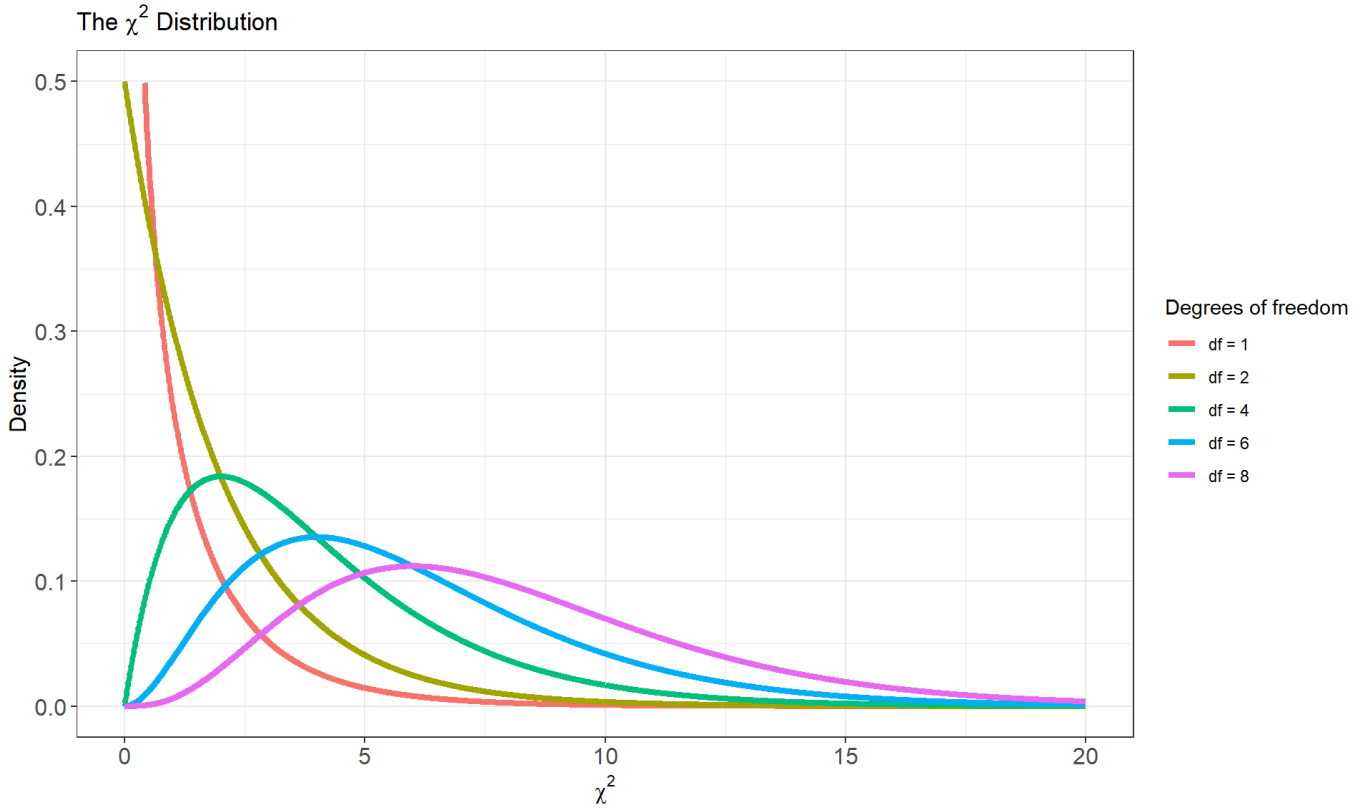
$$X^2 = \frac{(6022 - 6017.25)^2}{6017.25} + \frac{(2001 - 2005.75)^2}{2005.75}$$

$$= 0.0037 + 0.0112$$

$$\approx 0.015$$

The Chi-square distribution

To convert the above statistic into a p-value we need to know its sampling distribution. The statistic X^2 is called a χ^2 ("chi-squared") test statistic because its sampling distribution follows a special type of probability distribution called the χ^2 distribution. Like the t distribution, the χ^2 distributions form a family of distributions controlled by a single parameter - the degrees of freedom. We will use the notation $\chi^2(df)$ to depict a specific member of this family. For the goodness of fit test, the test statistic X^2 follows a χ^2 distribution with $k - 1$ degrees of freedom where k represents the number of categories of the qualitative variable



* Note from the graph above that as the degrees of freedom increase the χ^2 distribution converges in shape to a normal distribution.

- Under the special case of a $\chi^2(1)$ the χ^2 statistic is equal to the square of a standard normal distribution Z^2.
- In general, a χ^2 random variable with k degrees of freedom can be represented as a sum of k squared normal random variables

$$\chi^2_k = \sum_{i=1}^{k} Z_i^2 = Z_1^2 + Z_2^2 + \cdots Z_k^2$$

- Like the t distribution, we can look up the probability of different chi-squared random variables using the degrees of freedom and a table of pre-computed probabilities:

Upper tail	0.25	0.2	0.15	0.125	0.1	0.075	0.05	0.025	0.005	0.0025	5e-04
DF											
1	1.323	1.642	2.072	2.354	2.706	3.17	3.841	5.024	7.879	9.141	12.116
2	2.773	3.219	3.794	4.159	4.605	5.181	5.991	7.378	10.597	11.983	15.202
3	4.108	4.642	5.317	5.739	6.251	6.905	7.815	9.348	12.838	14.32	17.73
4	5.385	5.989	6.745	7.214	7.779	8.496	9.488	11.143	14.86	16.424	19.997
5	6.626	7.289	8.115	8.625	9.236	10.008	11.07	12.833	16.75	18.386	22.105
6	7.841	8.558	9.446	9.992	10.645	11.466	12.592	14.449	18.548	20.249	24.103
7	9.037	9.803	10.748	11.326	12.017	12.883	14.067	16.013	20.278	22.04	26.018
8	10.219	11.03	12.027	12.636	13.362	14.27	15.507	17.535	21.955	23.774	27.868

Upper tail	0.25	0.2	0.15	0.125	0.1	0.075	0.05	0.025	0.005	0.0025	5e-04
9	11.389	12.242	13.288	13.926	14.684	15.631	16.919	19.023	23.589	25.462	29.666
10	12.549	13.442	14.534	15.198	15.987	16.971	18.307	20.483	25.188	27.112	31.42
11	13.701	14.631	15.767	16.457	17.275	18.294	19.675	21.92	26.757	28.729	33.137
12	14.845	15.812	16.989	17.703	18.549	19.602	21.026	23.337	28.3	30.318	34.821
13	15.984	16.985	18.202	18.939	19.812	20.897	22.362	24.736	29.819	31.883	36.478
14	17.117	18.151	19.406	20.166	21.064	22.18	23.685	26.119	31.319	33.426	38.109
15	18.245	19.311	20.603	21.384	22.307	23.452	24.996	27.488	32.801	34.95	39.719
16	19.369	20.465	21.793	22.595	23.542	24.716	26.296	28.845	34.267	36.456	41.308
17	20.489	21.615	22.977	23.799	24.769	25.97	27.587	30.191	35.718	37.946	42.879
18	21.605	22.76	24.155	24.997	25.989	27.218	28.869	31.526	37.156	39.422	44.434
19	22.718	23.9	25.329	26.189	27.204	28.458	30.144	32.852	38.582	40.885	45.973
20	23.828	25.038	26.498	27.376	28.412	29.692	31.41	34.17	39.997	42.336	47.498
21	24.935	26.171	27.662	28.559	29.615	30.92	32.671	35.479	41.401	43.775	49.011
22	26.039	27.301	28.822	29.737	30.813	32.142	33.924	36.781	42.796	45.204	50.511
23	27.141	28.429	29.979	30.911	32.007	33.36	35.172	38.076	44.181	46.623	52
24	28.241	29.553	31.132	32.081	33.196	34.572	36.415	39.364	45.559	48.034	53.479
25	29.339	30.675	32.282	33.247	34.382	35.78	37.652	40.646	46.928	49.435	54.947
26	30.435	31.795	33.429	34.41	35.563	36.984	38.885	41.923	48.29	50.829	56.407
27	31.528	32.912	34.574	35.57	36.741	38.184	40.113	43.195	49.645	52.215	57.858
28	32.62	34.027	35.715	36.727	37.916	39.38	41.337	44.461	50.993	53.594	59.3
29	33.711	35.139	36.854	37.881	39.087	40.573	42.557	45.722	52.336	54.967	60.735
30	34.8	36.25	37.99	39.033	40.256	41.762	43.773	46.979	53.672	56.332	62.162
40	45.616	47.269	49.244	50.424	51.805	53.501	55.758	59.342	66.766	69.699	76.095
50	56.334	58.164	60.346	61.647	63.167	65.03	67.505	71.42	79.49	82.664	89.561
100	109.141	111.667	114.659	116.433	118.498	121.017	124.342	129.561	140.169	144.293	153.167
Z-score	0.674	0.842	1.036	1.15	1.282	1.44	1.645	1.96	2.576	2.807	3.291
Confidence Level:	50%	60%	70%	75%	80%	85%	90%	95%	99%	99.5%	99.9%

Lecture 29

Warm Up: χ^2 Goodness of Fit Test - Suppose you're hired by a dog food company to help them test three new dog food flavors. You recruit a random sample of 75 dogs and offer each dog a choice between the three flavors by placing bowls in front of them. Assuming that the dogs have no preference, you expect that the flavors will be equally popular among the dogs, with about 25 dogs choosing each flavor. After weeks of hard work, your dog food experiment is complete and you compile your data in a table:

Flavor	Observed Count	Expected Count
Garlic Blast	22	
Bacon Delight	30	
Chicken Crunch	23	

Conduct a χ^2 goodness of fit test to determine if the dogs show any preference for the three flavors being tested

Note that the GOF test doesn't tell us anything about the range of plausible values for the frequency parameters.

Measuring statistical association in two categorical variables

As stated previously, the goodness of fit test is a one-sample procedure for comparing the observed distribution of a single categorical variable to a specific model that we hypothesize under the null. We will now explore another type of inference which is also based on the χ^2 ("chi-square") distribution. This procedure is used to determine if two categorical variables share an association by exploring how the distribution of counts of one variable changes as we condition on the categories of another variable.

χ^2 test of independence

The χ^2 test of independence is used to examine the relationship between two categorical variables in a two-way table (also called a **contingency table**). Consider the following hypothetical table which compares the categorical variable A, which has two categories, with the categorical variable B which has three categories.

		Variable A		
		Category 1	Category 2	Marginal Distribution B
Variable B	Category 1	$P(\text{Category } 1A \mid \text{Category } 1B)$	$P(\text{Category } 2A \mid \text{Category } 1B)$	$\sum_j P(\text{Category } jA \mid \text{Category } 1B)$
	Category 2	$P(\text{Category } 1A \mid \text{Category } 2B)$	$P(\text{Category } 2A \mid \text{Category } 2B)$	$\sum_j P(\text{Category } jA \mid \text{Category } 2B)$
	Category 3	$P(\text{Category } 1A \mid \text{Category } 3B)$	$P(\text{Category } 2A \mid \text{Category } 3B)$	$\sum_j P(\text{Category } jA \mid \text{Category } 3B)$
	Marginal Distribution A	$\sum_i P(\text{Category } 1A \mid \text{Category } iB)$	$\sum_i P(\text{Category } 2A \mid \text{Category } iB)$	1

The two-way table above helps us to break down the relationship between the outcomes of *Variable A* and the outcomes of *Variable B*. As we can see from the table, the cells give the conditional probabilities (or more often frequencies/counts) of each category of A across each category of B. There are also usually row and column totals displayed in a two-way table which summarize the marginal distributions of each variable.

A χ**-square test of independence** takes advantage of the two-way table above to determine if the conditional distribution of the response variable (the columns) changes across the categories of the explanatory variable (the rows). In the example above, we can see that each row gives the conditional distribution of the response variable for a given category of the explanatory variable. If *variable A* and *variable B* are independent then the rows of the table should be more or less the same across all categories of the explanatory variable. If the outcomes of *variable A* depend on the outcomes of variable B, then the rows of the table will change as we move across the categories of *variable B*.

The null and alternative hypothesis of a χ-square test of independence are simple. The null hypothesis is that the two variables being tested are independent and the alternative is that they not independent

$$H_0 : \text{Variable A and Variable B are independent}$$

$$H_A : \text{Variable A and Variable B are NOT independent}$$

Like the goodness of fit test, the alternative hypothesis does not specify any particular direction. Because the two-way table can contain all sorts of possible associations, the alternative hypothesis cannot be described as one or two sided.

To test the null hypothesis of "no dependence" between two variables in a table with r rows and c columns, we take a similar approach to the goodness of fit test and compare the observed cell counts in all cells of the table to their expected cell counts under the null hypothesis.

In essence we are comparing two tables which represent two distributions: the table of **observed cell counts** which represents the observed joint distribution of the response and explanatory variables and the table of **expected cell counts** which represents expected joint distribution of the two variables the null hypothesis of no association.

The test statistic summarizes the differences (i.e distance) between the counts in the observed table with the counts in the expected table.

$$X^2_{obs} = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

Where o_{ij} represents the observed count for the ith row and j^{th} column of the contingency table:

The $r \times c$ table of observed counts

X.	Category.1	Category.2	X..1	Category.c	Total
Category 1	o_{11}	o_{12}	$\cdots$	o_{1c}	$\sum_{j=1}^{c} o_{1j}$
Category 2	o_{21}	o_{22}	$\cdots$	o_{2c}	$\sum_{j=1}^{c} o_{2j}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
Category r	o_{r1}	o_{r2}	$\cdots$	o_{rc}	$\sum_{j=1}^{c} o_{rj}$
Total	$\sum_{i=1}^{r} o_{i1}$	$\sum_{i=1}^{r} o_{i2}$	$\cdots$	$\sum_{i=1}^{r} o_{ic}$	$n = \sum_{i=1}^{r}\sum_{j=1}^{c} o_{ij}$

The expected counts for i, j^{th} cell of the table are computed from the observed counts such that

$$e_{ij} = \frac{\left[\sum_{i=1}^{r} o_{ij}\right] \times \left[\sum_{j=1}^{c} o_{ij}\right]}{\sum_{i=1}^{r}\sum_{j=1}^{c} o_{ij}} = \frac{\text{row total} \times \text{column total}}{n}$$

The test statistic follows a χ^2 distribution with $(r-1) \times (c-1)$ degrees of freedom, where r is the total number of row categories and c is the total column categories. Let's try it out on an example:

Example: Happiness and Income - As part of their annual survey, the General Social Survey explored the relationship between income and happiness. Happiness was recorded as "Not too happy", "Pretty happy", or "Very happy", and incomes were bracketed into three categories: "Below average", "Average", and "Above average". The counts are summarized in the table below:

The $r \times c$ table of observed counts

Income	Not Too Happy	Pretty Happy	Very Happy	Total
Above Average	29	178	135	342
Average	83	494	277	854
Below Average	104	314	119	537
Total	216	986	531	1733

Conduct a χ^2 test of independence to determine if there is evidence to support that income is related to happiness

- The table of expected counts is given below

The $r \times c$ table of expected counts

	Not Too Happy	Pretty Happy	Very Happy
Above Average	$\frac{342 \times 216}{1733} = 42.63$	$\frac{342 \times 986}{1733} = 194.58$	$\frac{342 \times 531}{1733} = 104.79$
Average	$\frac{854 \times 216}{1733} = 106.44$	$\frac{854 \times 986}{1733} = 485.89$	$\frac{854 \times 531}{1733} = 261.67$
Below Average	$\frac{537 \times 216}{1733} = 66.93$	$\frac{537 \times 986}{1733} = 305.53$	$\frac{537 \times 531}{1733} = 164.54$

- the table below gives the squared deviations across all cell counts

The $r \times c$ table of differences

	Not Too Happy	Pretty Happy	Very Happy
Above Average	$\frac{(42.63-29)^2}{42.63} = 4.36$	$\frac{(194.58-178)^2}{194.58} = 1.41$	$\frac{(104.79-135)^2}{104.79} = 8.71$
Average	$\frac{(106.44-83)^2}{106.44} = 5.16$	$\frac{(485.89-494)^2}{485.89} = 0.14$	$\frac{(261.67-277)^2}{261.67} = 0.9$
Below Average	$\frac{(66.93-104)^2}{66.93} = 20.53$	$\frac{(305.53-314)^2}{305.53} = 0.23$	$\frac{(164.54-119)^2}{164.54} = 12.6$

The χ^2 test statistic can be computed as the sum of the elements in the table above

$$X^2_{obs} = 4.36 + 5.16 + \cdots + 0.23 + 12.6 = 54.04$$

which follows a χ^2 distribution with $(3-1) \times (3-1) = 4$ degrees of freedom. The p-value is the probability of observing a value more extreme than $X^2_{obs} = 54.04$ which is

$$P(\chi^2_4 \geq 54.04 | H_0 \text{ True}) \approx 0$$

Thus, at $\alpha = 0.05$ level we would conclude that there is significant evidence that income and happiness are related.

References